

Advanced Algorithms — Calculation-Style Practice Questions

Multiple-choice (A–D). Designed for quick calculations (parameters, probabilities, asymptotics).

Questions

Q1 [Divide & Conquer Recurrence — Easy] Let $T(n)=2T(n/2)+n$ ($T(1)=1$). What is $T(n)$ in Θ -notation?

- A. $\Theta(\log n)$
- B. $\Theta(n)$
- C. $\Theta(n \log n)$
- D. $\Theta(n^2)$

Q2 [Divide & Conquer Recurrence — Easy] Let $T(n)=T(n/2)+1$ ($T(1)=1$). What is $T(n)$ in Θ -notation?

- A. $\Theta(1)$
- B. $\Theta(\log n)$
- C. $\Theta(n)$
- D. $\Theta(n \log n)$

Q3 [Divide & Conquer Recurrence — Medium] Let $T(n)=3T(n/2)+n$. What is $T(n)$ in Θ -notation?

- A. $\Theta(n)$
- B. $\Theta(n \log n)$
- C. $\Theta(n^{\log_2 3})$
- D. $\Theta(n^2)$

Q4 [Dynamic Programming (Coin Row) — Easy] Coin Row: coins=[6,7,1,30,8,2,4]. Max sum with no adjacent coins?

- A. 37
- B. 39
- C. 41
- D. 43

Q5 [0-1 Knapsack — Medium] 0-1 Knapsack capacity=6. Items (w,v): A(2,6), B(2,10), C(3,12). Max value?

- A. 16
- B. 18
- C. 22
- D. 28

Q6 [Fractional Knapsack — Medium] Fractional knapsack capacity=50. Items A(10,60), B(20,100), C(30,120). Greedy by v/w gives total value:

- A. 160

- B. 180
- C. 200
- D. 240

Q7 [MST (Kruskal) — Medium] Kruskal: sorted edges (A,B,1),(B,C,2),(A,C,3),(C,D,4). After adding first two edges, what about (A,C,3)?

- A. Add it (it's light).
- B. Skip it (it creates a cycle).
- C. Replace (B,C,2).
- D. Add only if D is isolated.

Q8 [MST (Prim) — Medium] Prim starts at A. Current tree has {A,B}. Crossing edges: (B,C,5),(A,C,2),(B,D,1). Which edge is chosen next?

- A. (B,C,5)
- B. (A,C,2)
- C. (B,D,1)
- D. Depends only on labels

Q9 [Reservoir Sampling — Hard] Reservoir sampling with $k=1$. Probability that item #5 remains in the sample after $n=20$ items?

- A. $1/5$
- B. $1/20$
- C. $5/20$
- D. $(1/5) \cdot (5/20) = 1/20$

Q10 [Reservoir Sampling — Hard] Reservoir sampling with $k=3$ over $n=12$. What is $P(\text{item \#4 is in final sample})$?

- A. $3/4$
- B. $3/12$
- C. $4/12$
- D. $1/12$

Q11 [Coupon Collector — Medium] Coupon collector: $E[\text{draws}] \approx n \cdot H_n$. With $n=10$ and $H_{10} \approx 2.93$, estimate E :

- A. ~ 10
- B. ~ 19.3
- C. ~ 29.3
- D. ~ 93

Q12 [Birthday Paradox — Medium] Birthday paradox: about how many samples until collisions likely in space $N=10,000$? (order-of-magnitude)

- A. ~ 10
- B. ~ 100

C. ~ 1

D. 000

Q13 [Probabilistic Counters — Medium] Probabilistic counter with base $a=2$. If counter is $k=4$, what is $P(\text{increment on next event})$?

A. $1/4$

B. $1/8$

C. $1/16$

D. $1/32$

Q14 [Probabilistic Counters — Hard] Base $a=2$ counter outputs $k=6$. Using estimator $(a^k - a + 1)/(a-1)$, estimated count is:

A. 63

B. 64

C. 61

D. 62

Q15 [Bloom Filters — Medium] Bloom filter $m=10,000$ bits, $n=1,000$ items. $k_{\text{opt}} \approx (m/n) \ln 2$. Closest k ?

A. ~ 3

B. ~ 7

C. ~ 14

D. ~ 70

Q16 [Bloom Filters — Hard] Bloom filter $m=10,000, n=1,000, k=7$. $p \approx (1 - e^{-kn/m})^k$. Using $e^{-0.7} \approx 0.50$, $p \approx$

A. $(0.5)^7 \approx 0.0078$

B. $(0.5)^7 \approx 0.50$

C. $(1-0.5)^7 \approx 0.50$

D. $(1-0.5)^7 \approx 0.0078$

Q17 [Count-Min Sketch — Medium] Count-Min width $w \approx \text{ceil}(e/\epsilon)$. If $\epsilon=0.02$ and $e \approx 2.718$, $w \approx$

A. ~ 50

B. ~ 100

C. ~ 136

D. ~ 272

Q18 [Count-Min Sketch — Medium] Count-Min depth $d \approx \text{ceil}(\ln(1/\delta))$. If $\delta=0.01$, $d \approx$

A. ~ 2

B. ~ 3

C. ~ 5

D. ~ 10

Q19 [Count-Min Sketch — Easy] Count-Min query: hashed counters across rows are [12,9,15,10]. Returned estimate is:

- A. 9
- B. 10
- C. 11.5
- D. 15

Q20 [Count-Min Sketch — Medium] True $\text{freq}(x)=8$ but Count-Min returns 11. Most likely reason:

- A. Underflow in counters.
- B. Hash collisions inflated counters.
- C. Min operator forces underestimation.
- D. Count-Min is exact.

Q21 [Misra–Gries — Hard] Misra–Gries with $k=3$ (max 2 counters). Stream: A,B,A,C,A,B,A. Which keys remain as candidates?

- A. {A}
- B. {B}
- C. {A,B}
- D. {B,C}

Q22 [Misra–Gries — Medium] Misra–Gries guarantee: any item with frequency $> n/k$ appears as candidate. If $n=1000, k=10$ threshold is:

- A. >10
- B. >50
- C. >100
- D. >500

Q23 [Distinct Counting — Hard] For a uniformly random binary hash output, $P(\text{trailing_zeros} \geq 3)$ equals:

- A. $1/2$
- B. $1/4$
- C. $1/8$
- D. $1/16$

Q24 [Distinct Counting — Medium] FM/HLL intuition: if max trailing zeros observed in a register is $R=5$, distinct count scale is about:

- A. ~ 5
- B. ~ 16
- C. ~ 32
- D. ~ 64

Q25 [Randomized Quicksort — Medium] If expected work is $\Theta(n \log^2 n)$, for $n=1024$, $n \cdot \log^2(n)$ is:

- A. ~ 1
- B. 024
- C. ~ 5

Answer Key + Brief Explanations

Q#	Correct
1	C
2	B
3	C
4	C
5	C
6	D
7	B
8	C
9	B
10	B
11	C
12	~10
13	C
14	A
15	B
16	A
17	C
18	C
19	A
20	B
21	C
22	C
23	C
24	C
25	~10

Q1 — Correct: C

Master theorem: $a=2, b=2 \Rightarrow n^{\log_b a}=n$; $f(n)=n \Rightarrow \Theta(n \log n)$.

Q2 — Correct: B

Each step halves $n \Rightarrow$ about $\log_2(n)$ steps.

Q3 — Correct: C

$n^{\log_2 3}$ dominates because $f(n)=n$ is smaller.

Q4 — Correct: C

Best is $7+30+4=41$ (no adjacent selections).

Q5 — Correct: C

Best feasible is B+C: weight 5, value 22.

Q6 — Correct: D

Take A,B fully (160). Remaining 20: take 20/30 of C $\Rightarrow$ +80, total 240.

Q7 — Correct: B

A and C already connected via A-B-C; adding A-C creates a cycle.

Q8 — Correct: C

Prim picks the minimum crossing edge: (B,D,1).

Q9 — Correct: B

With $k=1$, each item ends with probability $1/n \Rightarrow 1/20$.

Q10 — Correct: B

Each item has probability $k/n \Rightarrow 3/12$.

Q11 — Correct: C

$10 \times 2.93 \approx 29.3$.

Q12 — Correct: ~10

000

Q13 — Correct: C

Rule: increment with probability $1/a^k = 1/2^4 = 1/16$.

Q14 — Correct: A

$(2^6 - 2 + 1)/(1) = 64 - 1 = 63$.

Q15 — Correct: B

$m/n=10 \Rightarrow k \approx 10 \times 0.693 \approx 6.93 \rightarrow \sim 7$.

Q16 — Correct: A

$kn/m=0.7 \Rightarrow e^{-0.7} \approx 0.5 \Rightarrow p \approx (1-0.5)^7 = (0.5)^7 \approx 0.0078$.

Q17 — Correct: C

$w \approx \text{ceil}(2.718/0.02) = \text{ceil}(135.9) = 136$.

Q18 — Correct: C

$\ln(100) \approx 4.605 \Rightarrow d \approx 5$.

Q19 — Correct: A

Standard query is min across rows $\Rightarrow 9$.

Q20 — Correct: B

Collisions only increase counters $\rightarrow$ upward bias; min reduces but doesn't eliminate.

Q21 — Correct: C

With 2 counters, A and B remain as candidates; C triggers decrement but doesn't dominate.

Q22 — Correct: C

$n/k=100 \Rightarrow$ items with frequency >100 must appear as candidates.

Q23 — Correct: C

Need last 3 bits all 0 $\Rightarrow (1/2)^3=1/8$.

Q24 — Correct: C

$R \approx \log_2(D) \Rightarrow D \approx 2^5 \approx 32$.

Q25 — Correct: ~10

240